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**Passive Investing and the Rise of Mega-Firms**

## Contents

# 1 Big picture

- **Question.** How does the rise of passive investing affect asset prices across firms, and in particular why might the effect be much stronger for the very largest firms than for the rest of the market?
- **Core theoretical answer.** Passive flows disproportionately raise the prices of the economy's largest firms, especially when those firms are also in high demand by noise traders. The key object is not only *systematic* risk, but also the *idiosyncratic supply* of large firms that active investors must absorb.
- **Main mechanism.** When investors move from active to passive funds, passive funds buy firms in proportion to index weights. For a mega-firm, the extra demand can be accommodated only if experts hold a larger position — often a larger short position if noise traders already over-demand the stock. Because the firm is large, that extra position meaningfully raises experts' exposure to the firm's idiosyncratic risk. The stock price must therefore rise. Since the higher price also magnifies dollar idiosyncratic risk, the effect is amplified.
- **Why this is different from standard stories.** The paper is not about asymmetric information, and it is not just an "index inclusion" story. The mechanism survives even when the tracked index includes *all* firms and even when information is symmetric.
- **Main comparative-static result.** In the benchmark case where the index is the full market portfolio and noise traders are absent, a *pure switch* from active to passive leaves prices unchanged because experts and non-experts hold the same portfolio. But when passive flows reflect entry into equities, or when the index is partial, or when noise traders create demand imbalances, the largest firms move the most.
- **Main quantitative result from the model.** In the calibration, passive flows raise the prices of the largest firms much more than those of small firms. With a large-stock index and noise traders, even a pure switch from active to passive can raise the aggregate market because the gains of high-demand mega-firms dominate the losses elsewhere.
- **Main empirical evidence.** In the data, quarterly flows into S&P500 index funds predict relative outperformance of the largest S&P500 firms and higher idiosyncratic volatility for those firms. By contrast, the largest firms inside the S&P600 do *not* outperform following flows into that small-stock index.
- **Why the paper matters.** The paper implies that passive investing compresses financing costs primarily for mega-firms, can increase market concentration, and may bias the aggregate market toward overvaluation even without any net new money entering stocks.

# 2 Data and measurement (model inputs, calibration, and empirical data)

## 2.1 Model environment and key objects

- **Assets.** There is a riskless asset and $N$ stocks. Stock $n$ pays a dividend flow with a constant part, a systematic part, and an idiosyncratic part.
- **Agents.** *Experts* can hold all stocks without constraints and stand in for active investors. *Non-experts* can hold only the riskless asset and a capitalization-weighted index, and stand in

for passive investors.

- **Noise traders.** The model can include noise traders who hold a fixed number of shares of each stock. Their presence is important because it can force experts to hold underweights or even short positions in the most over-demanded large stocks.
- **Index structure.** The index can include all firms or only a subset. This allows the paper to study both all-market indexing and large-stock indexing such as the S&P500.
- **Key supply concept.** For any stock, the relevant object is the number of shares that experts must absorb after subtracting passive-fund demand and noise-trader demand. The model calls this "idiosyncratic supply" for each stock and "systematic supply" after aggregating across stocks with factor loadings.

## 2.2 Calibration design

- **Firm-size distribution.** The calibration uses about 1,686 firms in five size groups, chosen to mimic the empirical power-law size distribution of firms.
- **Group sizes.** Size group 5 has 6 firms, group 4 has 30, group 3 has 150, group 2 has 750, and group 1 has 750. Shares outstanding scale as $625\eta, 125\eta, 25\eta, 5\eta, \eta$, so the largest firms are economically dominant.
- **Active and passive masses.** The starting point is $(\mu_1, \mu_2) = (0.9, 0.1)$, where $\mu_1$ is the mass of experts and $\mu_2$ the mass of non-experts. Passive growth is modeled by increasing $\mu_2$ to 0.6.
- **Entry versus switching.** The paper distinguishes $\phi = 0$, where passive growth is entirely new entry into stocks, from $\phi = 1$, where passive growth is entirely a switch from active to passive and total investor mass stays fixed.
- **Index cases.** The calibration studies three cases: an all-stock index, a large-stock index containing only the top three size groups, and a small-stock index containing only the bottom three size groups.
- **Noise-trader cases.** In the baseline there are no noise traders. In the heterogeneous-demand case, half the stocks in each size group have low demand and half have high demand equal to 30% of issued shares.
- **Technology cases.** The calibration studies a *constant-*$b_n$ case, where systematic dividend exposure is the same for all stocks, and a *varying-*$b_n$ case, where systematic exposure declines with size so the model matches the empirical negative relation between size and CAPM beta.

## 2.3 Empirical data

- **Returns and market capitalization.** From CRSP.
- **Index composition.** S&P500 composition comes from CRSP, and S&P600 composition from Siblis Research.
- **Passive-fund assets.** Net assets for S&P500 index mutual funds come from ICI; ETF assets and S&P600 index-fund assets come from CRSP.
- **ETF sample.** The S&P500 ETF sample consists mainly of SPY, IVV, and VOO — essentially the plain-vanilla S&P500 ETF market.

- **Samples.** The S&P500 sample runs from 1996Q2 to 2020Q4. The S&P600 sample runs from 2001Q4 to 2020Q4.
- **Large-stock portfolios.** Within the S&P500 the paper studies top-10, top-50, top-100, top-150, and top-200 portfolios by lagged market capitalization. For the S&P600 it studies the top 60 firms, which correspond to the top decile of that index.
- **Idiosyncratic volatility.** Measured as the quarterly standard deviation of daily residual returns from a Fama-French three-factor regression.

### 2.4 Passive-flow measurement

$$\text{PassiveFlow}_{500,t} = \frac{\$\,\text{S\&P500 index assets}_t}{\$\,\text{S\&P500 index cap}_t} - \frac{\$\,\text{S\&P500 index assets}_{t-1}}{\$\,\text{S\&P500 index cap}_{t-1}}. \tag{1}$$

- **Interpretation.** This is the quarterly change in passive ownership of the index, scaled by index capitalization.
- **Magnitude.** In the S&P500 sample, the mean quarterly flow is 0.05% and the standard deviation is 0.09%. Cumulated over the sample, passive ownership rises by about 4.95% of index market capitalization.
- **Standardization.** In the regressions, the paper standardizes the passive-flow variable to mean zero and standard deviation one, so coefficients can be read as the effect of a one-standard-deviation increase in passive flows.

## 3 Models and empirical specifications (all equations + interpretation)

### 3.1 Dividend process and return definitions

$$D_{nt} = \bar{D}_n + b_n D_t^s + D_{nt}^i. \tag{2}$$

$$dD_t^s = \kappa^s(\bar{D}^s - D_t^s)dt + \sigma^s \sqrt{D_t^s}\, dB_t^s, \tag{3}$$

$$dD_{nt}^i = \kappa_n^i(\bar{D}_n^i - D_{nt}^i)dt + \sigma_n^i \sqrt{D_{nt}^i}\, dB_{nt}^i. \tag{4}$$

- **Interpretation.** Each firm's cash flow has a common macro component and a firm-specific component, both modeled as square-root processes.
- **Why square-root matters.** Dividend volatility rises with the level of dividends, so when prices rise, absolute idiosyncratic price fluctuations rise too. That is the ingredient that creates the paper's amplification mechanism.
- **Normalization.** The long-run mean of the systematic factor is set to one, and each firm's long-run mean dividend flow per share is normalized to one.

$$dR_{nt}^{sh} \equiv D_{nt}dt + dS_{nt} - rS_{nt}dt, \qquad dR_{nt} \equiv \frac{dR_{nt}^{sh}}{S_{nt}}. \tag{5}$$

- **Interpretation.** $dR^{sh}_{nt}$ is the share excess return and $dR_{nt}$ is the dollar excess return.

### 3.2 Portfolio constraints and mean-variance objectives

Experts and non-experts maximize mean-variance objectives over infinitesimal changes in wealth:

$$\max\ \mathbb{E}_t[dW_{1t}] - \frac{\rho}{2}\mathrm{Var}_t(dW_{1t}), \qquad \max\ \mathbb{E}[dW_{2t}] - \frac{\rho}{2}\mathrm{Var}(dW_{2t}). \tag{6}$$

$$z_{2nt} = \lambda\eta'_n, \qquad \mu_1 z_{1nt} + \mu_2\lambda\eta'_n + u_n = \eta_n. \tag{7}$$

- **Interpretation.** Non-experts can only hold the index in scaled form, so their stock positions are pinned down by index weights $\eta'_n$. Experts absorb the residual supply left over after passive demand and noise-trader demand.
- **Economic meaning of market clearing.** The more passive investors buy an included stock, the smaller the supply experts must hold. If the stock is large and volatile, that change can materially alter its discount rate.

### 3.3 Equilibrium price decomposition

The paper looks for affine pricing functions:

$$S_{nt} = \bar{S}_n + b_n S^s(D^s_t) + S^i_n(D^i_{nt}). \tag{8}$$

In equilibrium, this becomes

$$S_{nt} = \frac{\bar{D}_n}{r} + b_n a^s_1\left(\frac{\kappa^s}{r} + D^s_t\right) + a^i_{n1}\left(\frac{\kappa^i_n}{r}\bar{D}^i_n + D^i_{nt}\right). \tag{9}$$

Define

$$\Xi^s \equiv \left(\sum_{m=1}^{N}\frac{\eta_m - \mu_2\lambda\eta'_m - u_m}{\mu_1}b_m\right)(\sigma^s)^2, \qquad \Xi^i_n \equiv \frac{\eta_n - \mu_2\lambda\eta'_n - u_n}{\mu_1}(\sigma^i_n)^2. \tag{10}$$

Then

$$a^s_1 = \frac{2}{r + \kappa^s + \sqrt{(r+\kappa^s)^2 + 4\rho\,\Xi^s}}, \qquad a^i_{n1} = \frac{2}{r + \kappa^i_n + \sqrt{(r+\kappa^i_n)^2 + 4\rho\,\Xi^i_n}}. \tag{11}$$

- **Interpretation of $a^s_1$ and $a^i_{n1}$.** These coefficients are inverse discount-rate objects. The first prices systematic dividends; the second prices idiosyncratic dividends.
- **Systematic supply.** $\Xi^s$ is the risk-adjusted amount of aggregate systematic exposure experts must hold.
- **Idiosyncratic supply.** $\Xi^i_n$ is the risk-adjusted quantity of stock $n$'s idiosyncratic risk experts must hold.

- **Core pricing insight.** Lower idiosyncratic supply lowers the discount rate on idiosyncratic dividends. This matters little for small firms, whose idiosyncratic component is almost diversifiable at the market level, but it matters a lot for mega-firms because their firm-specific cash-flow risk is large in dollar terms.
- **Why volatility rises with price.** Because discounting is multiplicative rather than additive, a fall in supply raises both the price level and the absolute size of price movements.

### 3.4 Comparative statics of passive flows

Passive flows correspond to an increase in the mass $\mu_2$ of non-experts. The paper lets $\mu_1$ fall by a fraction $\phi \in [0,1]$ of that increase:

- $\phi = 0$: passive growth is entry by new investors into equities.
- $\phi = 1$: passive growth is a pure switch from active to passive.

The main comparative-static equation is

$$\frac{1}{S_{nt}}\frac{dS_{nt}}{d\mu_2} = \frac{\rho}{\mu_1 S_{nt}}\left[b_n\left(\sum_{m=1}^{N} 1_m b_m\right)(\sigma^s a_1^s)^2\left(\frac{\kappa^s}{r} + D_t^s\right)F^s + 1_n(\sigma_n^i a_{n1}^i)^2\left(\frac{\kappa_n^i}{r}\bar{D}_n^i + D_{nt}^i\right)F_n^i\right], \tag{12}$$

where

$$1_n \equiv -\mu_1 \frac{d}{d\mu_2}\left(\frac{\eta_n - \mu_2\lambda\eta_n' - u_n}{\mu_1}\right) \tag{13}$$

and $F^s, F_n^i$ are positive inverse-square-root terms that make prices more sensitive when discount rates are lower.

- **Interpretation.** Passive flows affect prices through two channels. The first is the standard market-beta channel. The second is the stock-specific idiosyncratic-supply channel, which is the paper's main innovation.
- **Key result in the benchmark case.** If the index is the whole market and noise traders hold the same fraction of every stock, then a pure switch from active to passive does not move prices: experts and non-experts hold the same portfolio.
- **Entry case.** If passive growth reflects entry into stocks, prices rise, and the rise is stronger for higher-beta stocks. But for very large stocks the rise is even larger than beta alone predicts because the idiosyncratic-supply term matters.

### 3.5 Why large stocks move more than CAPM beta predicts

In the benchmark all-market case, the paper rewrites the price effect as

$$\frac{1}{S_n}\frac{\partial S_n}{\partial(-\mathrm{MRP})} = \frac{\beta_n}{r + \beta^s \mathrm{MRP}} + w_n^i \beta_n^i \left[\frac{1}{r + \beta_n^i \mathrm{MRP}} - \frac{1}{r + \beta^s \mathrm{MRP}}\right]. \tag{14}$$

- **Interpretation.** The first term is ordinary CAPM logic: lower market risk premia raise prices in proportion to beta. The second term is the mega-firm term: if a stock has non-negligible idiosyncratic beta, passive flows raise its price more than its CAPM beta alone would suggest.

- **Small versus large firms.** For small stocks, $\beta_n^i \approx 0$, so the second term vanishes. For large stocks, $\beta_n^i$ is non-negligible because the stock itself is a large piece of the market portfolio.
- **Discount-rate intuition.** Idiosyncratic dividends are discounted at a lower rate than systematic dividends. So a given drop in the relevant discount rate has a bigger present-value effect on the idiosyncratic component.

A large stock out-reacts a smaller stock with similar beta when

$$\left(\frac{\kappa_n^i}{r}\bar{D}_n^i + D_{nt}^i\right) F_n^i > \left(\frac{\kappa^s}{r} D_t^s + 1\right) D_{nt}^i F^s. \tag{15}$$

- **Meaning of the condition.** The idiosyncratic-dividend present value must be sufficiently sensitive to passive flows relative to the systematic-dividend present value.

### 3.6 Empirical specifications

The main return-predictability regression is

$$R_{p,t}^{Exc} = \alpha + \beta \,\widehat{\text{PassiveFlow}}_{500,t} + \Gamma' X_t + \varepsilon_t, \tag{16}$$

where $R_{p,t}^{Exc}$ is the excess return of a large-stock portfolio within the S&P500 relative to the index, $\widehat{\text{PassiveFlow}}_{500,t}$ is standardized passive flow, and $X_t$ contains the S&P500 return, its lag, and $VIX$.

- **Interpretation of $\beta$.** A positive $\beta$ means that quarters with large passive inflows are quarters when the largest firms outperform the index.

The idiosyncratic-volatility panel regression is

$$\log(Vol_{i,t}^{\text{Idio}}) = \alpha_i + \delta_t + \beta_1 \,\widehat{\text{PassiveFlow}}_{t-1} \times \text{Top50}_i + \beta_2 \,\widehat{\text{PassiveFlow}}_{t-1} + \Gamma' Z_{i,t-1} + \varepsilon_{it}, \tag{17}$$

where $\text{Top50}_i$ indicates whether a firm belongs to the top 50 of the S&P500.

- **Interpretation of $\beta_1$.** If positive, passive flows raise idiosyncratic volatility more for the very largest firms, exactly as the theory predicts.

## 4 Identification and instruments (what makes the estimates credible)

### 4.1 This is not an IV paper

- The paper's strongest claims are theoretical and structural: they come from solving the equilibrium and tracing comparative statics.
- The empirical section is best read as *validation* and *discipline* for the model, not as a fully causal reduced-form design with an external instrument.
- So the right standard is: do the patterns in the data line up with the model's sharpest cross-sectional predictions?

### 4.2 Why the theory is sharp

- **Exact decomposition of passive growth.** The paper distinguishes new money into equities from reallocation from active to passive via the single parameter $\phi$. That is conceptually important because the model predicts dramatically different aggregate effects across those cases.
- **Cross-sectional testable implications.** The model does not merely say "passive flows move prices." It says that the strongest effects should appear for the economy's largest firms, especially if they are included in the tracked index and especially if they are also in high demand by noise traders.
- **Placebo implication.** The largest stocks inside a small-stock index should not show the same pattern, because they are not large in the economy-wide sense.

### 4.3 Why the empirical design is informative

- **Within-index relative returns.** The paper studies excess returns of large-stock portfolios relative to the index itself. That strips out common market-wide movements and focuses on the cross-sectional tilt toward mega-firms.
- **Controls.** The return regressions control for the contemporaneous and lagged index return and for $VIX$, which helps guard against the interpretation that passive flows are just proxying for broad market conditions.
- **Volatility test.** The idiosyncratic-volatility regressions with firm fixed effects and, alternatively, time fixed effects are especially useful because the theory predicts not just price changes but also a specific risk consequence for large firms.
- **S&P600 placebo.** The lack of a similar effect in the S&P600 is a powerful cross-check. It lines up exactly with the model's distinction between "largest in the index" and "largest in the economy."

### 4.4 Limits and how to read the evidence

- Passive flows may still be endogenous to broader investor demand conditions, so the empirical section should not be read as proving a clean causal reduced form.
- The empirical evidence is strongest when read jointly: mega-firm relative returns in the S&P500, larger idiosyncratic-volatility responses for top firms, and no analogous return effect for top firms inside the S&P600.
- The appendix robustness exercises — concentration regressions and a beginning-of-month flow proxy — further support the mechanism, but the paper's main contribution remains the theoretical one.

## 5 Tables and figures

The main text is table-heavy and contains no standalone main-text figures. The core exhibits are Tables 4.1–4.6 and 5.1–5.4.

## 5.1 Table 4.1: Calibrated price and return moments

**Read:**

- In the **constant-$b_n$** case, expected excess returns and CAPM betas rise with firm size: the largest group has beta 1.24 versus about 1.01 for the smallest groups.
- In the **varying-$b_n$** case, the calibration matches the empirical negative size-beta relation: beta falls from 1.38 for the smallest group to 0.96 for the largest.
- The table matters because it shows that the model can reproduce either benchmark CAPM-like intuition or the empirically more relevant negative size-beta pattern before studying passive flows.

**Table 4.1**
**Price and return moments for $(\mu_1, \mu_2) = (0.9, 0.1)$ and the baseline**

*A. Constant-$b_n$ case*

| Size group | Price | Expected excess return (%) | Return volatility (%) | CAPM beta | CAPM $R^2$ (%) |
|---|---|---|---|---|---|
| 1 (smallest) | 12.93 | 3.89 | 12.56 | 1.01 | 26.18 |
| 2 | 12.91 | 3.90 | 12.56 | 1.01 | 26.28 |
| 3 | 12.83 | 3.93 | 12.56 | 1.02 | 26.81 |
| 4 | 12.46 | 4.07 | 12.53 | 1.07 | 29.32 |
| 5 (largest) | 11.24 | 4.62 | 12.45 | 1.24 | 39.78 |

*B. Varying-$b_n$ case*

| Size group | Price | Expected excess return (%) | Return volatility (%) | CAPM beta | CAPM $R^2$ (%) |
|---|---|---|---|---|---|
| 1 (smallest) | 10.42 | 5.34 | 12.51 | 1.38 | 28.82 |
| 2 | 11.54 | 4.65 | 11.09 | 1.19 | 27.46 |
| 3 | 12.59 | 4.11 | 9.93 | 1.04 | 26.41 |
| 4 | 13.35 | 3.76 | 8.88 | 0.96 | 27.52 |
| 5 (largest) | 13.37 | 3.72 | 7.64 | 0.96 | 37.04 |

**Table 4.2**
**Percentage price change caused by passive flows in the baseline**

| | Entry into the stock market | | Switch from active to passive | |
|---|---|---|---|---|
| Size group | Constant-$b_n$ | Varying-$b_n$ | Constant-$b_n$ | Varying-$b_n$ |
| 1 (smallest) | 7.03 | 7.67 | 0 | 0 |
| 2 | 7.08 | 6.63 | 0 | 0 |
| 3 | 7.34 | 5.99 | 0 | 0 |
| 4 | 8.45 | 6.17 | 0 | 0 |
| 5 (largest) | 11.78 | 7.53 | 0 | 0 |

Table 4.1 shows the unconditional average of the price and the unconditional return moments for $(\mu_1, \mu_2) = (0.9, 0.1)$ and the baseline, in the constant-$b_n$ case (panel A) and the varying-$b_n$ case (panel B). When moving from the smallest to the largest size group, expected excess return and CAPM beta rise in the constant-$b_n$ case but decline in the varying-$b_n$ case. Stocks in size group 5 have the largest CAPM $R^2$, even in the varying-$b_n$ case where their CAPM beta is the smallest. This is because Proposition 2.1 implies that stock prices are less sensitive to idiosyncratic dividend shocks when idiosyncratic supply is large.

## 5.2 Table 4.2: Baseline passive-flow price effects

**Read:**

- When passive growth is due to **entry into stocks**, prices rise for all firms. In the constant-$b_n$ case the effect rises monotonically from 7.03% for the smallest group to 11.78% for the largest.
- In the varying-$b_n$ case the pattern is **U-shaped**: the effect declines from 7.67% to 5.99% for small and mid-sized firms, then rises again to 7.53% for the largest firms. This is the key "size beats beta" result.
- When passive growth is a **pure switch from active to passive**, all price changes are zero in the all-stock benchmark. That is the clean CAPM benchmark result.

## 5.3 Table 4.3: Partial-index counterfactuals

**Read:**

- If the index contains only the **top three size groups**, a pure switch from active to passive raises the price of the largest group by 5.98% in the constant-$b_n$ case and 5.32% in the varying-$b_n$ case.
- If the index contains only the **bottom three size groups**, the same pure switch causes the price of the largest economy-wide firms to *fall* sharply: −9.54% and −6.58%.
- This table is one of the paper's most important quantitative exhibits because it shows that index inclusion matters only for large stocks. It also shows how a partial index can make aggregate prices move even when passive growth is just a reallocation across active and passive investors.

**Table 4.3**
**Percentage price change caused by passive flows into a partial index**

*A. Index includes only top three size groups*

| | Entry into the stock market | | Switch from active to passive | |
|---|---|---|---|---|
| Size group | Constant-$b_n$ | Varying-$b_n$ | Constant-$b_n$ | Varying-$b_n$ |
| 1 (smallest) | 6.90 | 7.50 | −0.42 | −0.56 |
| 2 | 6.91 | 6.44 | −0.60 | −0.69 |
| 3 | 7.32 | 6.00 | 0.04 | 0.13 |
| 4 | 8.80 | 6.59 | 1.50 | 1.67 |
| 5 (largest) | 13.43 | 9.03 | 5.98 | 5.32 |

*B. Index includes only bottom three size groups*

| | Entry into the stock market | | Switch from active to passive | |
|---|---|---|---|---|
| Size group | Constant-$b_n$ | Varying-$b_n$ | Constant-$b_n$ | Varying-$b_n$ |
| 1 (smallest) | 7.02 | 7.66 | −0.02 | −0.01 |
| 2 | 7.11 | 6.67 | 0.15 | 0.17 |
| 3 | 7.56 | 6.21 | 0.97 | 0.93 |
| 4 | 7.29 | 5.09 | −4.12 | −3.64 |
| 5 (largest) | 8.13 | 4.80 | −9.54 | −6.58 |

**Table 4.4**
**Percentage price change caused by passive flows with noise traders**

| | | Entry into the stock market | | Switch from active to passive | |
|---|---|---|---|---|---|
| Size group | Noise-trader demand | Constant-$b_n$ | Varying-$b_n$ | Constant-$b_n$ | Varying-$b_n$ |
| 1 (smallest) | Low | 7.27 | 7.95 | −0.02 | −0.02 |
| | High | 7.27 | 7.95 | −0.00 | −0.01 |
| 2 | Low | 7.32 | 6.89 | −0.04 | −0.05 |
| | High | 7.31 | 6.89 | 0.02 | 0.03 |
| 3 | Low | 7.54 | 6.20 | −0.17 | −0.18 |
| | High | 7.52 | 6.18 | 0.15 | 0.16 |
| 4 | Low | 8.49 | 6.22 | −0.68 | −0.63 |
| | High | 8.47 | 6.25 | 0.73 | 0.70 |
| 5 (largest) | Low | 11.31 | 7.24 | −1.90 | −1.38 |
| | High | 11.71 | 7.78 | 2.59 | 2.03 |

## 5.4 Table 4.4: Noise-trader heterogeneity

**Read:**

- Noise-trader demand barely matters for small stocks, but it matters a lot for the largest firms.
- Under a pure switch from active to passive with an all-stock index, the largest low-demand firms fall by −1.90% and −1.38%, while the largest high-demand firms rise by 2.59% and 2.03%.
- That asymmetry is the model's main amplification mechanism in quantitative form: passive flows bite hardest when experts already have to lean against noise-trader demand.

## 5.5 Table 4.5: Large-stock index plus noise traders

**Read:**

- This is the strongest quantitative version of the paper's mechanism.
- With a large-stock index and noise traders, a pure active-to-passive switch raises the price of the largest *high-demand* firms by 10.03% in the constant-$b_n$ case and 9.58% in the varying-$b_n$ case.
- The corresponding effect for the largest low-demand firms is only about 2%. So the skew toward mega-firms is highly asymmetric.
- This table explains why the aggregate market can rise even when no new money enters stocks: the gain for high-demand mega-firms dominates the rest.

**Table 4.5**
**Percentage price change caused by passive flows into a partial index with noise traders**

| | | Entry into the stock market | | Switch from active to passive | |
|---|---|---|---|---|---|
| Size group | Noise-trader demand | Constant-$b_n$ | Varying-$b_n$ | Constant-$b_n$ | Varying-$b_n$ |
| 1 (smallest) | Low | 7.15 | 7.80 | −0.43 | −0.59 |
| | High | 7.15 | 7.79 | −0.42 | −0.57 |
| 2 | Low | 7.16 | 6.72 | −0.60 | −0.71 |
| | High | 7.16 | 6.71 | −0.54 | −0.64 |
| 3 | Low | 7.51 | 6.19 | −0.21 | −0.16 |
| | High | 7.49 | 6.18 | 0.13 | 0.21 |
| 4 | Low | 8.76 | 6.55 | 0.40 | 0.57 |
| | High | 8.77 | 6.61 | 2.08 | 2.28 |
| 5 (largest) | Low | 12.56 | 8.37 | 2.05 | 1.99 |
| | High | 13.37 | 9.39 | 10.03 | 9.58 |

**Table 4.6**
**Change in idiosyncratic volatility caused by passive flows from active**

| | | All-stock index | | Large-stock index | |
|---|---|---|---|---|---|
| Size group | Noise-trader demand | Constant-$b_n$ | Varying-$b_n$ | Constant-$b_n$ | Varying-$b_n$ |
| 1 (smallest) | Low | 0.00 | 0.00 | 0.06 | 0.09 |
| | High | 0.00 | 0.00 | 0.06 | 0.09 |
| 2 | Low | 0.00 | 0.00 | 0.05 | 0.06 |
| | High | 0.00 | 0.00 | 0.05 | 0.06 |
| 3 | Low | 0.00 | −0.01 | 0.07 | 0.07 |
| | High | 0.00 | 0.01 | 0.08 | 0.10 |
| 4 | Low | −0.03 | −0.05 | 0.10 | 0.12 |
| | High | 0.03 | 0.05 | 0.16 | 0.24 |
| 5 (largest) | Low | −0.08 | −0.12 | 0.18 | 0.24 |
| | High | 0.11 | 0.16 | 0.43 | 0.73 |

## 5.6 Table 4.6: Idiosyncratic-volatility effects

**Read:**

- Following a pure switch from active to passive, idiosyncratic volatility rises much more for the largest firms, especially when those firms are high-demand and in a large-stock index.

- For size group 5 high-demand stocks, the volatility change is 0.11 and 0.16 in the all-stock-index case, but jumps to 0.43 and 0.73 in the large-stock-index case.
- This is critical because the paper's mechanism is not just about prices. It is explicitly about the idiosyncratic risk that experts must absorb.

## 5.7 Table 5.1: Descriptive statistics

**Read:**

- Mean quarterly passive flow into the S&P500 is 0.05% with a standard deviation of 0.09%.
- Mean excess returns of large-stock portfolios relative to the S&P500 are slightly negative on average, but the dispersion is substantial: for example, the top-50 value-weighted excess-return series has a standard deviation of 1.86% per quarter.
- For all S&P500 firms, mean log idiosyncratic volatility is −4.28, with meaningful cross-sectional variation.
- The table shows the raw objects that the empirical section is trying to connect: index flows, mega-firm relative returns, and firm-level idiosyncratic volatility.

**Table 5.1**
**Descriptive statistics**

A. Aggregate Variables

| | Mean | Std. dev. | 25th pctl. | 50th pctl. | 75th pctl. | Skewness | Exc. kurt. | N |
|---|---|---|---|---|---|---|---|---|
| $R^{Exc}_{Top10,SP500}$ | −0.02 | 3.86 | −2.04 | −0.45 | 2.59 | −0.04 | 0.39 | 99 |
| $R^{Exc}_{Top50EW,SP500}$ | −0.15 | 1.56 | −1.07 | −0.39 | 0.72 | 0.51 | 0.95 | 99 |
| $R^{Exc}_{Top50,SP500}$ | −0.17 | 1.86 | −1.27 | −0.20 | 1.09 | −0.16 | 0.50 | 99 |
| $R^{Exc}_{Top100,SP500}$ | −0.17 | 1.28 | −0.91 | −0.23 | 0.60 | −0.61 | 2.39 | 99 |
| $R^{Exc}_{Top150,SP500}$ | −0.19 | 0.95 | −0.69 | −0.19 | 0.42 | −0.49 | 1.76 | 99 |
| $R^{Exc}_{Top200,SP500}$ | −0.17 | 0.78 | −0.61 | −0.19 | 0.37 | −0.47 | 1.28 | 99 |
| $R_{SP500}$ | 2.68 | 8.50 | −0.77 | 3.41 | 7.60 | −0.56 | 0.55 | 99 |
| $PassiveFlow_{SP500}$ | 0.05 | 0.09 | 0.01 | 0.05 | 0.10 | 0.33 | 3.62 | 99 |
| $R^{Exc}_{Top60EW,SP600}$ | −0.05 | 3.41 | −1.30 | 0.08 | 1.55 | −0.04 | 2.49 | 77 |
| $R^{Exc}_{Top60,SP600}$ | −0.12 | 3.89 | −1.68 | 0.27 | 1.97 | −0.54 | 4.19 | 77 |
| $R_{SP600}$ | 3.14 | 9.97 | −0.32 | 3.99 | 8.17 | −0.63 | 1.35 | 77 |
| $PassiveFlow_{SP600}$ | 0.04 | 0.11 | −0.01 | 0.03 | 0.09 | −0.34 | 3.99 | 77 |
| $VIX$ | 20.36 | 7.59 | 14.57 | 19.31 | 24.92 | 1.80 | 6.03 | 99 |

B. Firm-Level Variables For All S&P500 Firms

| | Mean | Std. dev. | 25th pctl. | 50th pctl. | 75th pctl. | Skewness | Exc. kurt. | N |
|---|---|---|---|---|---|---|---|---|
| $\log(Vol_{Idio})$ | −4.28 | 0.50 | −4.63 | −4.31 | −3.95 | 0.35 | 0.40 | 46,831 |

**Table 5.2**
**Passive flows into the S&P500 and excess returns on S&P500 large-stock portfolios**

*A. Top 50 firms*

| Variables | $R^{Exc}_{Top50EW,SP500}$ | $R^{Exc}_{Top50,SP500}$ | $R^{Exc}_{Top50EW,SP500}$ | $R^{Exc}_{Top50,SP500}$ |
|---|---|---|---|---|
| $\widehat{PassiveFlow}_{SP500}$ | 0.00557 | 0.00553 | 0.00531 | 0.00528 |
| | (3.65) | (3.69) | (4.19) | (3.66) |
| Constant | −0.00150 | −0.00168 | −0.000253 | −0.00137 |
| | (−0.92) | (−0.80) | (−0.12) | (−0.54) |
| Observations | 99 | 99 | 99 | 99 |
| Controls | N | N | Y | Y |
| Adjusted $R^2$ | 0.127 | 0.088 | 0.210 | 0.125 |

*B. All large-stock portfolios*

| Variables | $R^{Exc}_{Top10,SP500}$ | $R^{Exc}_{Top50,SP500}$ | $R^{Exc}_{Top100,SP500}$ | $R^{Exc}_{Top150,SP500}$ | $R^{Exc}_{Top200,SP500}$ |
|---|---|---|---|---|---|
| $\widehat{PassiveFlow}_{SP500}$ | 0.00687 | 0.00528 | 0.00303 | 0.00208 | 0.00145 |
| | (2.46) | (3.66) | (3.02) | (2.28) | (1.64) |
| Constant | −0.00156 | −0.00137 | −0.00177 | −0.00189 | −0.00149 |
| | (−0.32) | (−0.54) | (−1.00) | (−1.40) | (−1.34) |
| Observations | 99 | 99 | 99 | 99 | 99 |
| Controls | Y | Y | Y | Y | Y |
| Adjusted $R^2$ | 0.049 | 0.125 | 0.105 | 0.124 | 0.125 |

## 5.8 Table 5.2: Passive flows and S&P500 large-stock outperformance

**Read:**

- A one-standard-deviation increase in S&P500 passive flows raises the quarterly excess return of the top-50 portfolio by about 0.528–0.557 percentage points, depending on weighting and controls.
- The effect is stronger the more one zooms into the very top of the size distribution: 0.145 points for the top 200, 0.208 for the top 150, 0.303 for the top 100, 0.528 for the top 50, and 0.687 for the top 10.
- The paper translates the top-50 estimate into a cumulative effect of about 29% over the sample, which is economically very large.

- This is the cleanest empirical counterpart to the model: quarters with high passive inflows are quarters when the biggest S&P500 firms outperform the index most.

### 5.9 Table 5.3: Passive flows and idiosyncratic volatility

**Read:**

- The interaction between lagged passive flow and the top-50 indicator is positive and significant: 19.32 or 18.46 depending on the fixed-effects specification.
- The paper converts this into a useful magnitude: a one-standard-deviation rise in passive flow increases idiosyncratic volatility by about 1.86% for non-top-50 firms and about 3.60% for top-50 firms.
- This is exactly the risk-side prediction of the theory. Passive flows are not just making large firms expensive; they are also making the idiosyncratic positions experts must absorb riskier.

**Table 5.3**
**Passive flows into the S&P500 and idiosyncratic return volatility of S&P500 stocks**

| | $\log(Vol_{Idio})$ | $\log(Vol_{Idio})$ |
|---|---|---|
| L.$PassiveFlow_{SP500} \times Top50$ | 19.32 | 18.46 |
| | (2.53) | (2.45) |
| L.$PassiveFlow_{SP500}$ | 20.63 | |
| | (1.21) | |
| L.$Top50$ | −0.0477 | −0.0671 |
| | (−2.86) | (−4.83) |
| Observations | 45,737 | 45,737 |
| Controls | Y | Y |
| Firm fixed effects | Y | Y |
| Time fixed effects | N | Y |
| Adjusted $R^2$ | 0.601 | 0.712 |

**Table 5.4**
**Passive flows into the S&P600 and excess returns on S&P600 large-stock portfolios**

| Variables | $R^{Exc}_{Top60EW,S\&P600}$ | $R^{Exc}_{Top60,S\&P600}$ | $R^{Exc}_{Top60EW,S\&P600}$ | $R^{Exc}_{Top60,S\&P600}$ |
|---|---|---|---|---|
| $\widehat{PassiveFlow}_{S\&P600}$ | −0.00541 | −0.00540 | −0.00397 | −0.00392 |
| | (−1.26) | (−1.04) | (−0.94) | (−0.75) |
| Constant | −0.000462 | −0.00121 | 0.00245 | 0.00161 |
| | (−0.12) | (−0.28) | (0.67) | (0.39) |
| Observations | 77 | 77 | 76 | 76 |
| Controls | N | N | Y | Y |
| Adjusted $R^2$ | 0.025 | 0.019 | 0.197 | 0.164 |

### 5.10 Table 5.4: S&P600 placebo test

**Read:**

- Repeating the return test for the top 60 firms in the S&P600 yields coefficients that are negative and statistically insignificant, roughly between −0.0054 and −0.0039.
- This is a sharp placebo. The largest firms inside a small-stock index are not mega-firms in the economy-wide sense, so the theory says they should not respond the same way.
- The null result is therefore supportive rather than disappointing: it helps isolate the paper's core prediction that what matters is being large in the economy, not merely large inside any arbitrary index.

## 6 Short summary

- **Conclusion.** Passive investing does not raise all stock prices equally. It tilts demand toward the very largest firms, and because active investors must absorb the residual idiosyncratic risk, the biggest firms move the most.
- **Intuition.** A mega-firm is too large for its firm-specific risk to be negligible. When passive funds buy it mechanically, the experts who take the other side need more compensation, which raises the stock's price and idiosyncratic volatility.

- **Empirical lesson.** The strongest effects of passive flows show up in the largest S&P500 firms and not in the largest firms of the S&P600. That is exactly the cross-sectional pattern the model predicts.

# 7 Conclusion

## 7.1 Core conclusion (what the paper establishes)

- **Main theoretical result.** Passive flows affect large firms not only through market beta but also through an idiosyncratic-supply channel.
- **Benchmark implication.** A pure active-to-passive switch need not change prices in a full-market CAPM benchmark.
- **Main non-benchmark implication.** Once one allows for partial indices or heterogeneous noise-trader demand, passive flows disproportionately raise the prices of included, high-demand mega-firms and can even raise the aggregate market under a pure switch.
- **Risk implication.** Passive flows also raise the idiosyncratic volatility of the largest firms.
- **Empirical confirmation.** The largest S&P500 firms outperform the index following passive inflows, and their idiosyncratic volatilities rise more. The corresponding effect is absent in the S&P600 placebo.

## 7.2 Economic intuition

- **Cap-weighted demand is inherently size-biased.** Passive funds allocate more dollars to larger firms by construction.
- **Mega-firms carry non-negligible idiosyncratic risk.** Because they are large, their firm-specific dividend shocks are not tiny relative to the aggregate market.
- **Experts face an amplification loop.** More passive demand pushes experts into larger residual positions, which raises required compensation, which raises price, which further scales up idiosyncratic dollar risk.
- **Noise-trader demand makes the effect asymmetric.** If a mega-firm is already over-demanded, passive flows can make active investors even more uncomfortable holding the offsetting position, producing especially large price effects.

## 7.3 Takeaway

This paper gives a clean and important theory of why passive investing can favor mega-firms even without any information frictions. Its main message is that capitalization-weighted indexing loads demand onto the firms whose idiosyncratic risk is least negligible, so passive flows reshape the cross-section of stock prices, not just the level of the market. The empirical evidence on S&P500 flows, large-firm relative returns, and idiosyncratic volatility lines up closely with that mechanism.